CAPSEL / INCA

PhD Implementation Roadmap

Complete phase-by-phase guide to implementation, experiments, paper submissions, and thesis completion

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36-Month Timeline · 2 Papers · 1 PhD

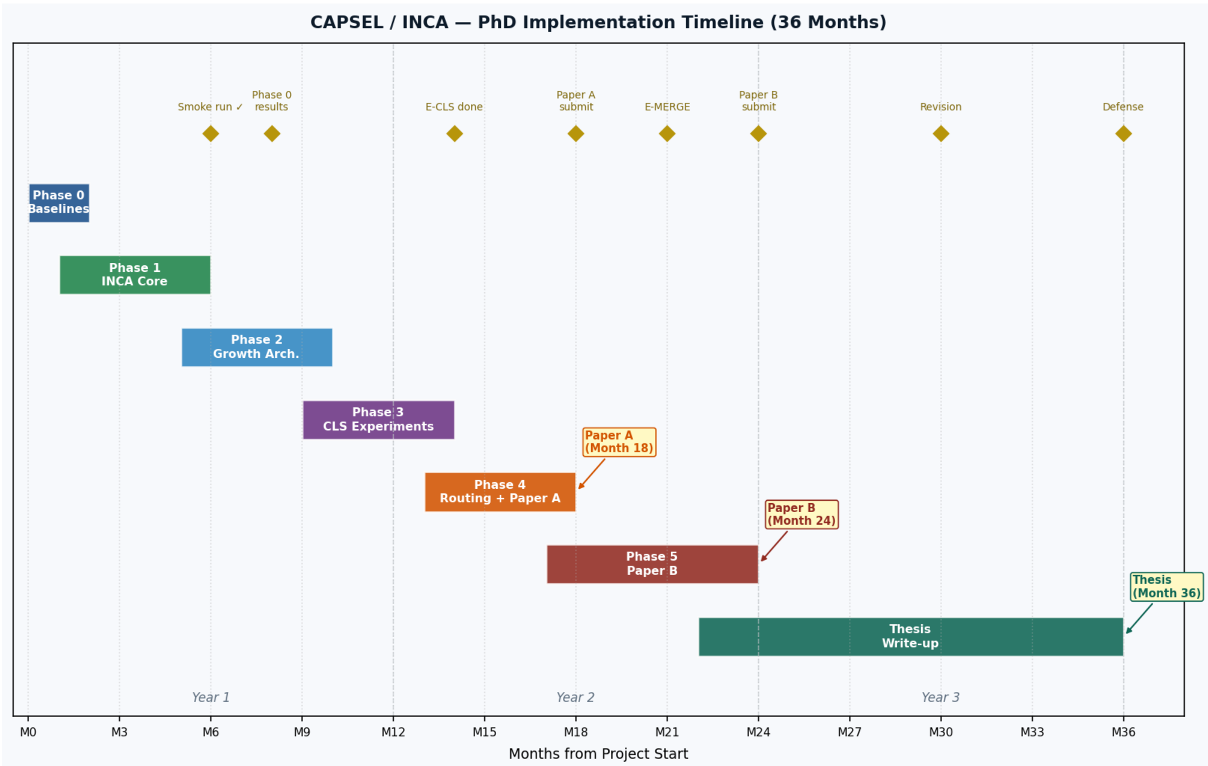
This document is your single implementation reference. It lays out every phase, every task, every experiment, and every deadline needed to go from today's working smoke run to a defended PhD thesis. Read it once end-to-end, then use it as a checklist. Nothing in the CAPSEL master PDF contradicts this — this roadmap *operationalises* everything described there.

■ End Goal (in one sentence)

Demonstrate that INCA's saturation-driven, grow-as-you-learn architecture achieves lower backward transfer and better forward transfer than all eight baselines on TiC-LM, publish two papers, and defend.

1. Master Timeline

The project runs across 36 months and produces two conference papers before the thesis. Each phase is colour-coded consistently throughout this document. Phases overlap deliberately — Phase 1 infrastructure work continues while Phase 0 baseline experiments are finishing.



Phase	Duration	Primary goal	Paper connection	Status
Phase 0 — Baselines	M0–M2	Run B1–B8 on CC-News; get numbers into baseline_report.md	Sets comparison floor for Paper A	■ In progress
Phase 1 — INCA Core	M1–M6	Multi-signal saturation detector; replay; growing block framework	Core contribution of Paper A	■ Smoke run ✓
Phase 2 — Growth Arch.	M5–M10	Lateral adapters; width expansion; G-chooser; ablation suite	Architecture ablations for Paper A §5	■ Not started
Phase 3 — CLS Exps	M9–M14	Frozen-block probes; representation drift; CKA analysis	Mechanistic analysis section Paper A §6	■ Not started
Phase 4 — Routing+Paper A	M13–M18	UCLBR router; full TiC-LM run; 3 seeds; Paper A write + submit	■ Paper A submitted Month 18	■ Not started
Phase 5 — Paper B	M17–M24	Sequential block merging; router recal.; post-merge bound; Paper B	■ Paper B submitted Month 24	■ Not started
Thesis Write-up	M22–M36	Thesis draft; revisions; defence preparation	Wraps both papers + new chapters	■ Not started

ASE 0 · BASELINES

Months 0 – 2 · Status: In Progres

Before training a single INCA block you need honest numbers for all eight baselines on the same data and the same evaluation protocol. Without this you cannot write Table 1 of Paper A. Phase 0 is pure engineering: make the eight baseline trainers run identically except for the continual-learning method, then evaluate BWT, FWT, and accuracy-at-end on CC-News.

Baselines B1–B8

ID	Task / Deliverable	Weeks	Status	Output file
B1	Naive fine-tune (no CL method)	W1–2	✓ Code done	train_naive.py
B2	Replay-only (uniform sampling)	W1–2	✓ Code done	train_replay.py
B3	EWC (Fisher diagonal, Soen 2025 bug fixes applied)	W1–2	✓ Code done	train_ewc.py
B4	L2P (Learning-to-Prompt)	W2–3	✓ Code done	train_l2p.py
B5	LoRA-MoE (8 experts, 4 active)	W2–3	✓ Code done	train_lora_moe.py
B6	LLaMA-Pro (block expansion, fixed schedule)	W3–4	✓ Code done	train_llama_pro.py
B7	PNN (Progressive Neural Networks)	W3–4	✓ Code done	train_pnn.py
B8	BlockStack (naive stacking, no saturation signal)	W4–5	✓ Code done	train_blockstack.py

Deliverables to close Phase 0

- Run all 8 baselines on **CC-News** (3 seeds each) — produces results/phase0/
- Fill **baselines_report.md** with BWT / FWT / Acc table
- Verify EWC Fisher computation uses Soen 2025 corrected diagonal (no double-counting)
- Compute **WER / perplexity** on held-out test split; export to CSV
- Commit Phase 0 results to git with tag phase0-results
- Write 1-page internal note: which baseline is hardest to beat and why

■■ Critical: what Phase 1 needs from Phase 0

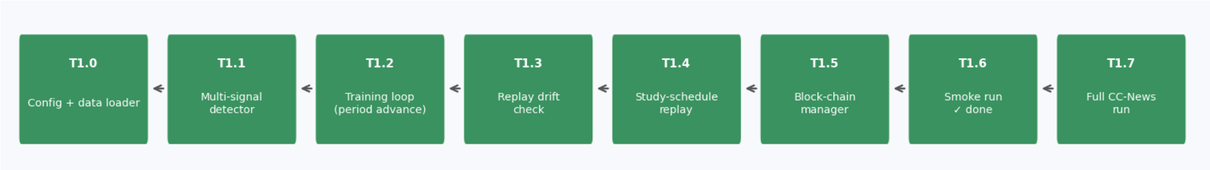
The exact cc_news_train / cc_news_val / cc_news_test splits used by Phase 0 must be frozen and reused unchanged by all INCA experiments. Any drift in data splits invalidates the comparison. Freeze splits before running Phase 1.

Save split manifest: results/phase0/data_split_manifest.json

IASE 1 · INCA CORE

Months 1 – 6 · Status: Smoke run co

Phase 1 builds the complete INCA training loop: the growing block chain (inca_layer_manager.py), the four-signal saturation consensus detector (inca_plateau.py), the study-schedule replay buffer (inca_replay.py), CKA monitoring (inca_cka.py), and the main training script (train_inca_v3.py). All of this is now code-complete and smoke-tested. Phase 1 is about verifying it works *correctly and robustly* on CC-News across multiple periods.



Completed tasks (smoke run ✓)

ID	Task / Deliverable	Weeks	Status	Output file
T1.0	INCAConfig dataclass + YAML loader	W1	✓ Done	inca_config.py
T1.1	GradNormTracker + RIR + LossPlateauDetector + CKAMonitor	W2	✓ Done	inca_plateau.py
T1.2	Training loop: period advance, BLOCK_FULL, PERIOD_LEARNED	W3	✓ Done	train_inca_v3.py
T1.3	Replay drift check (had_replay_before guard fixed)	W3	✓ Done	train_inca_v3.py
T1.4	StudyScheduleBuffer: N_revise + p_hard/p_easy/p_mid sampling	W2	✓ Done	inca_replay.py
T1.5	INCALayerManager: chain + embedding skip + inter-block projs	W2–3	✓ Done	inca_layer_manager.py
T1.6	Smoke YAML config + verified single-period training pass	W4	✓ Done	configs/smoke.yaml

Remaining Phase 1 tasks

ID	Task / Deliverable	Weeks	Status	Output file
T1.7	Full CC-News run: 6 periods, 3 seeds; collect BWT/FWT/Acc	W5–8	■ TODO	results/phase1/cc_news/
T1.8	SPRT-based saturation detector (replace fixed patience)	W5–6	■ TODO	inca_plateau.py
T1.9	Fisher-based structured pruning before freeze	W6–7	■ TODO	inca_fisher_prune.py
T1.10	Checkpoint save/load tested across grow events	W5	■ TODO	train_inca_v3.py
T1.11	Visualisation: block diagram from .pt file	W4	✓ Done	scripts/visualize_inca.py
T1.12	Compare INCA-v3 vs B6 (LLaMA-Pro) head-to-head on CC-News	W8	■ TODO	results/phase1/comparison.md

Bug fixes already applied (do not re-open)

- **gnorm=0.0 bug**: captured after zero_grad → fixed, last_grad_n captured before step
- **T1.3 false-fire**: had_replay_before flag missing → fixed, guard added
- **head_mask removed**: newer transformers API → fixed in INCABlock.forward()
- **cache_position=None**: T5Attention crash → fixed, arange passed to each layer
- **Parallel vs sequential**: memorandum specifies arch (a) → implemented with inter_block_projs

GROWTH ARCHITECTURE

Months 5 – 10 · Status: Not starte

Phase 1 only uses G-VERT (depth growth). Phase 2 adds the remaining two primitives and the automatic chooser that picks between them based on the saturation signal profile. This is where INCA becomes *adaptive* rather than just *growing*. Phase 2 produces the ablation experiments that fill Section 5 of Paper A.

New files to write

ID	Task / Deliverable	Weeks	Status	Output file
T2.1	inca_lateral.py — rank-r lateral adapters from each frozen block to current; α_k=0 init	W1–2	■	src/inca_lateral.py
T2.2	inca_width.py — Net2Net width expansion (G-HORIZ); function-preserving split	W2–3	■	src/inca_width.py
T2.3	inca_growth_chooser.py — G-VERT vs G-HORIZ vs G-LAT decision logic from saturation profile	W3–4	■	src/inca_growth_chooser.py
T2.4	Wire chooser into train_inca_v3.py — growth_mode: 'auto'|'vert'|'horiz'|'lat'	W4	■	train_inca_v3.py
T2.5	E-SCOPE: ablate lateral adapter rank r ∊ {4,8,16}; compare BWT/FWT	W5–6	■	results/phase2/e_scope/
T2.6	E-GROW: compare G-VERT / G-HORIZ / G-LAT / auto-chooser, 3 seeds	W6–8	■	results/phase2/e_grow/
T2.7	E-PRUNE: Fisher pruning sweep p ∊ {5,10,15,20}%; measure param count vs BWT	W7–8	■	results/phase2/e_prune/

Architecture (a) — sequential chain (already implemented)

The inter-block projection chain is already in inca_layer_manager.py. Each new block receives proj_i(prev_output) + embedding_hidden. Projections are identity-initialised (function-preserving). The incoming projection is *unfrozen* when its target block becomes trainable, allowing the new block to learn how to use frozen block output.

Architecture (c) — lateral adapters (Phase 2 target)

Lateral adapters are low-rank connections (rank r) from each frozen block directly to the current trainable block. Gate α_k=0 at init means the architecture is function-preserving at grow time: the new block starts identically to a G-VERT clone. As training proceeds, α_k can grow to allow the current block to selectively access older representations. This is E-SCOPE in the ablation table.

■ Phase 2 key decision point

After E-GROW results come in: if auto-chooser consistently outperforms any fixed primitive by >2% BWT, use auto-chooser for all Paper A experiments. If results are mixed, default to G-VERT only (simpler story for reviewers).

E 3 · CLS EXPERIMENTS

Months 9 – 14 · Status: Not starte

Phase 3 is the *mechanistic* phase. You have a working INCA model with frozen and trainable blocks. Now you probe what the frozen blocks have learned, measure representation drift via CKA, and build the narrative that frozen blocks act as a 'neocortex' while the replay buffer acts as 'hippocampus'. This section directly produces Paper A Section 6 (Analysis).

ID	Task / Deliverable	Weeks	Status	Output file
T3.1	probe_frozen_blocks.py — train lightweight probes on each block's frozen hidden states; report accuracy per block	W1–2	■	scripts/probe_frozen_blocks.py
T3.2	E-CLS1: verify frozen blocks retain early-period knowledge; accuracy should not decay on probes from period 0	W2–3	■	results/phase3/e_cls1/
T3.3	E-CLS2: CKA between block 0 and block N across training; confirm representation divergence increases with chain depth	W3–4	■	results/phase3/e_cls2/
T3.4	E-CLS3: replay buffer analysis — does hard-sample emphasis actually change BWT vs uniform replay?	W4–5	■	results/phase3/e_cls3/
T3.5	E-CLS4: grokking guard test — does MI progress floor prevent premature block freeze?	W5–6	■	results/phase3/e_cls4/
T3.6	E-CLS5: SPRT sensitivity — vary α/β; measure FP (premature grow) and FN (missed saturation) rates	W6–7	■	results/phase3/e_cls5/
T3.7	Write analysis narrative (≈2 pages) ready to paste into Paper A §6	W7–8	■	notes/cls_analysis_narrative.md

CLS framing — the story to tell in Paper A

- **Frozen blocks = neocortex**: stores compressed, stable knowledge from earlier periods
- **Replay buffer = hippocampus**: fast-access episodic store that prevents catastrophic forgetting
- **Freeze-and-grow = consolidation**: the moment a block freezes mirrors memory consolidation during sleep
- **CrossAttentionSelector = recall**: routes current task to the most relevant frozen representation
- This framing is optional in the main paper but very useful for the motivation section and related work

4 · ROUTING + PAPER A

Months 13 – 18 · Status: Not start

Phase 4 upgrades the router from CrossAttentionSelector to UCLBR, runs the full TiC-LM benchmark, and writes Paper A. This is the highest-stakes phase: everything before was building up to these numbers. Plan the GPU compute budget carefully — TiC-LM with 3 seeds × 9 methods (8 baselines + INCA) on flan-t5-base is roughly 40–60 GPU-hours.

ID	Task / Deliverable	Weeks	Status	Output file
T4.1	inca_uclbr.py — Read-ME pre-gating + DeepSeek aux-loss-free load balance + uncertainty-calibrated confidence routing	W1–3	■	src/inca_uclbr.py
T4.2	Wire --selector uclbr into train_inca_v3.py and validate on CC-News first	W3	■	train_inca_v3.py
T4.3	E-ROUTE: CrossAttentionSelector vs HebbianSelector vs UCLBR on CC-News; pick best for TiC-LM run	W3–4	■	results/phase4/e_route/
T4.4	E-SAT: full saturation ablation — any-1 / any-2 / any-3 / any-4 signal consensus; measure false-grow rate	W4–5	■	results/phase4/e_sat/
T4.5	TiC-LM Track A main run: INCA + B1–B8, 3 seeds, flan-t5-base, 12-month subset	W5–9	■	results/phase4/tic_lm_track_a/
T4.6	E-SCALE: flan-t5-large (780M) single-seed to verify scaling	W8–9	■	results/phase4/e_scale/
T4.7	Paper A write — target NeurIPS / ICLR / ICML main track	W9–12	■	paper_a/
T4.8	Paper A internal review + revisions	W11–12	■	paper_a/revision_v1/

Paper A — structure

Section	Title	Content source
§1	Introduction	Problem, gap, contribution bullet points
§2	Related Work	B1–B8 + LLaMA-Pro + TiC-LM survey; use reading list
§3	Method	INCA architecture (a); saturation detector; replay; SPRT
§4	Experiments	TiC-LM Track A Table 1; BWT/FWT/Acc vs all baselines
§5	Ablations	E-GROW / E-SCOPE / E-SAT / E-PRUNE / E-ROUTE
§6	Analysis	E-CLS1–5; frozen-block probes; CKA drift; CLS framing
§7	Conclusion	Summary; limitations (normalisation-by-count in selector); future (arch c)

■ Paper A submission deadline

Target: NeurIPS (deadline typically late May). If Phase 4 slips past Month 16, fall back to ICLR (deadline typically October). ICML (deadline typically January) is the final safety net. Do not submit to a workshop instead of the main track — the contribution is strong enough for main.

PAPER B — BLOCK MERGING

Months 17 – 24 · Status: Not start

Paper B addresses INCA's Achilles heel: the chain grows indefinitely. Sequential block merging (TIES-style) compresses redundant frozen blocks when their CKA similarity exceeds τ_merge = 0.90, keeping the model bounded in size while retaining knowledge. Paper B provides a formal post-merge router error bound (theoretical contribution) and demonstrates on TiC-LM Track B (Pythia-160M) that bounded-size INCA matches unbounded INCA within 1% BWT after 24 periods.

ID	Task / Deliverable	Weeks	Status	Output file
T5.1	inca_block_merge.py — TIES-style merge: trim, elect sign, disjoint merge, distillation refinement, router recalibration	W1–4	■	src/inca_block_merge.py
T5.2	Post-merge router error bound — derive and prove in supplementary	W2–5	■	paper_b/proof/
T5.3	E-MERGE: merge cadence K ∈ {2,3,5} grow events; measure accuracy drop and chain length	W4–6	■	results/phase5/e_merge/
T5.4	TiC-LM Track B main run: bounded-INCA vs unbounded-INCA vs SEEKR vs LoRA-MoE; Pythia-160M; 24 periods; 3 seeds	W5–9	■	results/phase5/tic_lm_track_b/
T5.5	E-TAU: merge threshold τ_merge ∈ {0.85, 0.90, 0.95}; accuracy vs compression tradeoff	W7–8	■	results/phase5/e_tau/
T5.6	Paper B write — target NeurIPS Efficient ML / ICML main track	W8–12	■	paper_b/
T5.7	Paper B internal review + revisions	W11–12	■	paper_b/revision_v1/

Paper B — structure

Section	Title	Content source
§1	Introduction	Problem: unbounded growth; contribution: bounded with guarantees
§2	Background	INCA (cite Paper A); TIES-merging; model merging survey
§3	Method	Sequential merging algorithm; distillation refinement; recalibration
§4	Theoretical Analysis	Post-merge router error bound; assumptions; proof sketch
§5	Experiments	TiC-LM Track B Table 1; E-MERGE; E-TAU
§6	Conclusion	Bounded continual LMs are feasible; future: task-free NN-CUSUM

E 0 · THESIS WRITE-UP

Months 22 – 36 · Status: Not start

The thesis is not just Papers A and B stapled together. UK/standard thesis format requires a coherent narrative with original introductory and concluding chapters, a full literature review chapter, and a methods chapter that unifies the two papers. Allow 6 months for writing and 2–4 months for examiner review and corrections.

Month	Thesis milestone	Notes
M22–M24	Chapter 1: Introduction (10–15 pp)	Write while Paper B is finishing
M22–M25	Chapter 2: Literature Review (30–40 pp)	Expand from Paper A §2; add Paper B related work
M24–M27	Chapter 3: INCA Architecture & Training	Paper A §3–§4 expanded with more derivation detail
M25–M28	Chapter 4: Growth Primitives & Ablations	Paper A §5–§6; add extra ablation runs if reviewers asked
M27–M30	Chapter 5: Bounded INCA (Paper B content)	Paper B §3–§5; add proof details in appendix
M28–M31	Chapter 6: Discussion & Future Work	CLS framing; UCLBR; task-free NN-CUSUM; open problems
M30–M32	Full draft → supervisor review	Expect 2–3 rounds of revisions
M32–M34	Corrections + polishing	Typography, figures, references
M34	Thesis submitted to examiners	Typically 6–8 weeks for viva scheduling
M36	Viva voce defence	Prepare 30-min presentation + 60-min Q&A

Thesis chapter outline

- **Ch 1: Introduction** — Motivation, research questions, contributions, thesis map
- **Ch 2: Literature Review** — Continual learning taxonomy; LLMs for CL; grow-then-prune methods
- **Ch 3: INCA: Architecture & Method** — Full technical description; training loop; saturation detection; replay
- **Ch 4: Experimental Evaluation (Paper A)** — TiC-LM results; all ablations; CLS analysis
- **Ch 5: Bounded INCA via Block Merging (Paper B)** — Merge algorithm; error bound; TiC-LM Track B results
- **Ch 6: Discussion** — Limitations; CLS framing; future directions (UCLBR, NN-CUSUM, arch c)
- **App A: Proofs** — SPRT formulation; post-merge router bound full derivation
- **App B: Hyperparameters** — Full config tables for all experiments
- **App C: Compute Budget** — GPU-hours breakdown; reproducibility statement

2. Complete Experiment Catalogue

Every experiment has a unique ID, a stated hypothesis, the phase it runs in, and which paper it feeds. Run experiments in phase order. Never run a TiC-LM main experiment before the CC-News pilot confirms the method works.

Exp ID	Name	Hypothesis tested	Phase	Paper
E-BASE	Baseline sweep	8 standard CL baselines give lower bound for BWT/FWT on CC-News	P0	A
E-SMOKE	Smoke run	INCA training loop runs without error for 1 period, 500 steps	P1	—
E-CCNEWS	CC-News pilot	INCA-v3 beats LLaMA-Pro (B6) on BWT; confirms method works before TiC-LM	P1	C
E-GROW	Growth primitives	Auto-chooser >= best fixed primitive (G-VERT/G-HORIZ/G-LAT) by ≥1% BWT	P2	A §5
E-SCOPE	Lateral rank	Rank 8 lateral adapters give best BWT/param tradeoff	P2	A §5
E-PRUNE	Fisher pruning	p=10-20% pruning reduces params 15% with <0.5% BWT loss	P2	A §5
E-CLS1	Frozen probes	Frozen block 0 retains ≥90% period-0 probe accuracy at end of training	P3	A §6
E-CLS2	CKA drift	CKA(block0, blockN) decreases monotonically as chain grows	P3	A §6
E-CLS3	Hard replay	p_hard=0.70 sampling reduces BWT by ≥2% vs uniform replay	P3	A §6
E-CLS4	Grokking guard	MI floor prevents at least 1 premature freeze across 6-period run	P3	A §6
E-CLS5	SPRT sensitivity	α=0.05 β=0.10 gives false-grow rate <5% on CC-News	P3	A §6
E-ROUTE	Router ablation	UCLBR >= CrossAttn >= Hebbian on CC-News BWT	P4	A §5
E-SAT	Consensus rule	Any-2 consensus minimises FP+FN vs any-1 or any-3	P4	A §5
E-TIC-A	TiC-LM Track A	INCA beats all 8 baselines on BWT; LLaMA-Pro is closest competitor	P4	A §4
E-SCALE	Scale test	flan-t5-large results hold: INCA gap vs LLaMA-Pro ≥1% BWT	P4	A §5
E-MERGE	Merge cadence	K=3 merges keeps chain ≤4 blocks while matching unbounded BWT within 1%	P5	B §5
E-TAU	Merge threshold	τ_merge=0.90 is sweet spot: compression ≥40% with <1% BWT regression	P5	B §5
E-TIC-B	TiC-LM Track B	Bounded-INCA matches unbounded within 1% BWT on Pythia-160M, 24 periods	P5	B §4

3. Paper Submission Guide

Paper A — Saturation-driven growing encoder for continual LM

Item	Target	Notes
Venue	NeurIPS	Primary. Deadline typically late May. 8 pages + refs.
Fallback 1	ICLR	Deadline typically early October. 8 pages.
Fallback 2	ICML	Deadline typically January. 8 pages.
Min. result	INCA > LLaMA-Pro on BWT by ≥1% absolute, 3 seeds, TiC-LM Track A	Non-negotiable
Good result	INCA in top-2 on FWT also; scale holds at flan-t5-large	Strengthens paper
Story	Saturation-driven grow > fixed-schedule grow > no-grow	Core narrative
Risk	TiC-LM run slips past M16 → fall to ICLR	Plan M14 pilot run

Paper B — Bounded-size continual LM via sequential block merging

Item	Target	Notes
Venue	NeurIPS Efficient ML Workshop / ICML	Paper B has a stronger efficiency angle
Fallback	ICLR or NeurIPS main	Elevate if post-merge bound is tight and novel
Min. result	Bounded-INCA matches unbounded within 1% BWT at K=3, chain ≤4 blocks	Non-negotiable
Good result	Formal error bound is tight (matches empirical within 0.5%)	Theory upgrade
Story	Practical bounded continual LM with guarantees	Cite Paper A; Paper A baseline needed first
Risk	Merging hurts BWT > 2% → investigate τ_merge / distillation hyperparams	

4. Compute Budget

All GPU estimates assume A100-40GB or equivalent. Adjust if using V100 (×1.8) or H100 (×0.55). The budget below is the *minimum viable* run — add 30% for debugging and re-runs.

Experiment	Model	Seeds	Est. GPU-h	Phase
E-BASE: 8 baselines × CC-News	flan-t5-base	3	~18 GPU-h	P0
E-CCNEWS: INCA pilot	flan-t5-base	1	~3 GPU-h	P1
E-GROW + E-SCOPE + E-PRUNE	flan-t5-base	3	~20 GPU-h	P2
E-CLS1–5	flan-t5-base	1	~8 GPU-h	P3
E-ROUTE + E-SAT	flan-t5-base	3	~10 GPU-h	P4
E-TIC-A: 9 methods × TiC-LM 12mo	flan-t5-base	3	~50 GPU-h	P4
E-SCALE: TiC-LM 12mo	flan-t5-large	1	~12 GPU-h	P4
E-MERGE + E-TAU	Pythia-160M	3	~12 GPU-h	P5
E-TIC-B: 4 methods × TiC-LM 24mo	Pythia-160M	3	~30 GPU-h	P5
Buffer (reruns, debugging)	—	—	~30 GPU-h	All
TOTAL	—	—	~193 GPU-h	—

■ Compute strategy

Request 200 GPU-hours from your supervisor / HPC allocation before starting Phase 4. Run E-BASE and E-CCNEWS first (≈21 GPU-h) — these are fast and validate the setup. Gate the TiC-LM main runs on a successful CC-News pilot.

Compressed TiC-LM option: use only 6 months of TiC-LM instead of 12. This halves Track A compute to ~25 GPU-h. Acceptable for submission if you note it clearly.

5. Week-by-Week Plan (Months 1–6)

This is the immediate action plan. Months 7–36 are governed by the phase tables above. Print this page and tick off each week.

Week	Month	Primary task	By end of week
W1	M1	Run B1–B8 on CC-News (Phase 0)	All 8 baselines running, no crashes
W2	M1	Collect Phase 0 results (3 seeds each)	baselines_report.md numbers filled in
W3	M2	Fix T1.7: full CC-News run INCA, 6 periods	train_inca_v3.py runs end-to-end 6 periods
W4	M2	SPRT detector (T1.8): replace fixed patience	inca_plateau.py uses SPRT thresholds
W5	M2	Fisher pruning (T1.9)	inca_fisher_prune.py integrated
W6	M3	Checkpoint save/load test (T1.10)	Resume from mid-training verified
W7	M3	E-CCNEWS pilot: INCA vs LLaMA-Pro CC-News	Preliminary BWT comparison in hand
W8	M3	E-CCNEWS final 3-seed run + analysis	results/phase1/comparison.md written
W9	M4	Phase 2 start: inca_lateral.py skeleton (T2.1)	Rank-r adapter forward pass working
W10	M4	inca_lateral.py complete + unit test	α_k=0 init verified function-preserving
W11	M5	inca_width.py Net2Net width expansion (T2.2)	G-HORIZ tested on 2-block chain
W12	M5	inca_growth_chooser.py (T2.3)	Chooser picks G-VERT vs G-HORIZ vs G-LAT
W13	M5	Wire chooser into train_inca_v3.py (T2.4)	growth_mode: auto|vert|horiz|lat working
W14	M6	E-SCOPE: lateral rank ablation	r $\in$ {4,8,16} results on CC-News
W15	M6	E-GROW: growth primitive ablation (3 seeds)	Auto vs fixed primitive BWT comparison
W16	M6	E-PRUNE: Fisher pruning sweep	p sweep results; pick best p for Paper A

6. Risk Register

Known risks, their impact level, and the contingency plan for each.

Risk	Likelihood	Impact	Contingency
INCA does not beat LLaMA-Pro on BWT	Medium	HIGH	Switch primary claim to 'saturation-driven growth converges faster'; show sam
TiC-LM run slips past Month 16	Medium	HIGH	Fall back to ICLR (Oct deadline); use CC-News results as main Table 1 with T
Post-merge BWT regression > 2%	Medium	MEDIUM	Increase distillation epochs; reduce merge frequency K; Paper B story become
GPU allocation insufficient (< 150 GPU-h)	Low	HIGH	Compressed TiC-LM (6 months); use Pythia-160M for all experiments; flan-t5-
Transformers API breaking changes	Medium	LOW	Pin transformers==4.40.0 in requirements.txt; test on each new version before
Paper A rejected at NeurIPS	Medium	LOW	Revise with reviewer feedback; resubmit ICLR. Rejection from main track ≠ ba
Thesis examiner requests major corrections	Low	MEDIUM	Build in 4-month corrections buffer (M34–M36 window); don't schedule defens

7. PhD Completion Checklist

Tick each item in order. You cannot skip an item — each one is a dependency for the items below it.

Phase 0 — Baselines (Must complete before Phase 1 full run)

- [ ] All 8 baseline trainers run without error on CC-News
- [ ] baselines_report.md filled with BWT / FWT / Acc for B1–B8
- [ ] Data split manifest frozen: results/phase0/data_split_manifest.json
- [ ] Git tag: phase0-results

Phase 1 — INCA Core (Must complete before Phase 2)

- [ ] Smoke run: single period, 500 steps, no errors ✓ DONE
- [ ] Full 6-period CC-News run with INCA (3 seeds)
- [ ] SPRT detector replaces fixed patience in inca_plateau.py
- [ ] Fisher pruning integrated and tested
- [ ] Checkpoint save/load verified across grow events
- [ ] E-CCNEWS: INCA vs LLaMA-Pro comparison written up
- [ ] Git tag: phase1-results

Phase 2 — Growth Architecture (Must complete before Paper A write)

- [ ] inca_lateral.py: rank-r lateral adapters, α_k=0 init
- [ ] inca_width.py: Net2Net G-HORIZ expansion
- [ ] inca_growth_chooser.py: auto-selection logic
- [ ] E-GROW, E-SCOPE, E-PRUNE results in hand
- [ ] Growth primitive for Paper A selected (auto-chooser or G-VERT only)
- [ ] Git tag: phase2-ablations

Phase 3 — CLS Experiments (Must complete before Paper A §6)

- [ ] probe_frozen_blocks.py written and validated
- [ ] E-CLS1–5 all run and results documented
- [ ] cls_analysis_narrative.md (2 pages) ready to paste into Paper A
- [ ] Git tag: phase3-cls

Phase 4 — Routing + Paper A (Month 13–18)

- [ ] inca_uclbr.py written and tested
- [ ] E-ROUTE: best router selected
- [ ] E-SAT: any-2 consensus validated
- [ ] E-TIC-A: full TiC-LM Track A run (3 seeds) complete
- [ ] E-SCALE: flan-t5-large single seed complete
- [ ] Paper A draft written (all sections)
- [ ] Paper A supervisor review ×2
- [ ] Paper A submitted to NeurIPS (or fallback)
- [ ] Git tag: paper-a-submission

Phase 5 — Paper B (Month 17–24)

- [ ] inca_block_merge.py: TIES merge + distillation + router recal.
- [ ] Post-merge error bound derived and written up
- [ ] E-MERGE, E-TAU, E-TIC-B results in hand
- [ ] Paper B draft written (all sections)
- [ ] Paper B supervisor review ×2
- [ ] Paper B submitted
- [ ] Git tag: paper-b-submission

Thesis (Month 22–36)

- [ ] All 6 chapters drafted
- [ ] Full draft to supervisor (M30)
- [ ] Two rounds of supervisor revisions
- [ ] Thesis submitted to examiners (M34)
- [ ] Viva preparation: 30-min talk + anticipated questions
- [ ] Viva passed / minor corrections completed (M36)
- [ ] ■ PhD awarded

8. Target File Tree (end of project)

Every file listed here must exist and be committed before you submit the thesis.

```
Phase0/
train_naive.py / train_replay.py / train_ewc.py
train_l2p.py / train_lora_moe.py / train_llama_pro.py
train_pnn.py / train_blockstack.py
results/phase0/baselines_report.md
results/phase0/data_split_manifest.json
```

```
Phase1/
src/
inca_config.py ← INCAConfig dataclass
inca_plateau.py ← multi-signal consensus + SPRT
inca_replay.py ← study-schedule replay buffer
inca_cka.py ← CKA reference set monitor
inca_selectors.py ← CrossAttentionSelector / Hebbian
inca_layer_manager.py ← block chain, sequential arch (a) ✓
inca_fisher_prune.py ← structured Fisher pruning
inca_uclbr.py ← UCLBR router (Phase 4)
scripts/
visualize_inca.py ← block diagram from .pt ✓
probe_frozen_blocks.py ← Phase 3 CLS probes
configs/
smoke.yaml ✓ cc_news.yaml tic_lm_a.yaml tic_lm_b.yaml
train_inca_v3.py ✓
```

```
Phase2/
src/
inca_lateral.py ← rank-r lateral adapters (arch c)
inca_width.py ← Net2Net G-HORIZ expansion
inca_growth_chooser.py ← G-VERT/HORIZ/LAT auto-chooser
```

```
results/
phase0/ phase1/ phase2/ phase3/ phase4/ phase5/
each: *.csv *.png *.md
```

```
paper_a/ paper_b/ thesis/
```

9. Final Note

The architecture is sound, the code is smoke-tested, and the hypothesis is falsifiable. The hardest part — deciding *what* to build — is done. What remains is execution: run the experiments in phase order, write up what you find honestly, and trust the process.

Three rules for the next 36 months

1. Never run a TiC-LM main experiment before the CC-News pilot confirms the method works.

2. Freeze data splits before Phase 1. Changing splits invalidates every comparison.

3. If INCA underperforms on a metric, report it and explain why — reviewers respect honesty more than cherry-picked results.

Good luck. The work is good.